The documentation and process conversion measures necessary to comply with this revision shall be completed by 17 August 1995

INCH-POUND

MIL-S-19500/290E
17 May 1995
SUPERSEDING
MIL-S-19500/290D
22 January 1992

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, SWITCHING
TYPES 2N2904, 2N2904A, 2N2904AL, 2N2905, 2N2905A, AND 2N2905AL,
JAN, JANTX, JANTXV, JANS, JANTXVM, JANTXVD, JANTXVR, JANTXVH,
JANSM, JANSD, JANSR, AND JANSH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for PNP, silicon, switching transistors. Four levels of product assurance are provided for each device type as specified in MIL-S-19500. Provision for radiation hardness assurance (RHA) to four radiation test levels is provided for JANTXV and JANS product assurance levels for type 2N2905A. RHA level designators "M", "D", "R", and "H" are appended to the device prefix to identify devices which have passed RHA requirements.

1.2 Physical dimensions. See 3.3 herein (TO-39).

1.3 Maximum ratings.

Types	P_T 1/ T_A = +25°C	P_T 2/ T_C = +25°C	V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_{STG} and T_J
	W	W	V dc	V dc	V dc	mA dc	°C
2N2904, 2N2905	0.6	3.0	60	40	5	600	-65 to +200
2N2904A, 2N2905A, 2N2904AL, 2N2905AL	0.6	3.0	60	60	5	600	-65 to +200

1/ Derate linearly, 3.43 mW/°C for T_A > +25°C.
2/ Derate linearly, 17.2 mW/°C for T_C > +25°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Research Laboratory, Electronics and Power Sources Directorate, ATTN: AMSRL-EP-RD, Ft. Monmouth, NJ 07703-5601 by using the Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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1.4 Primary electrical characteristics at T_A = +25°C.

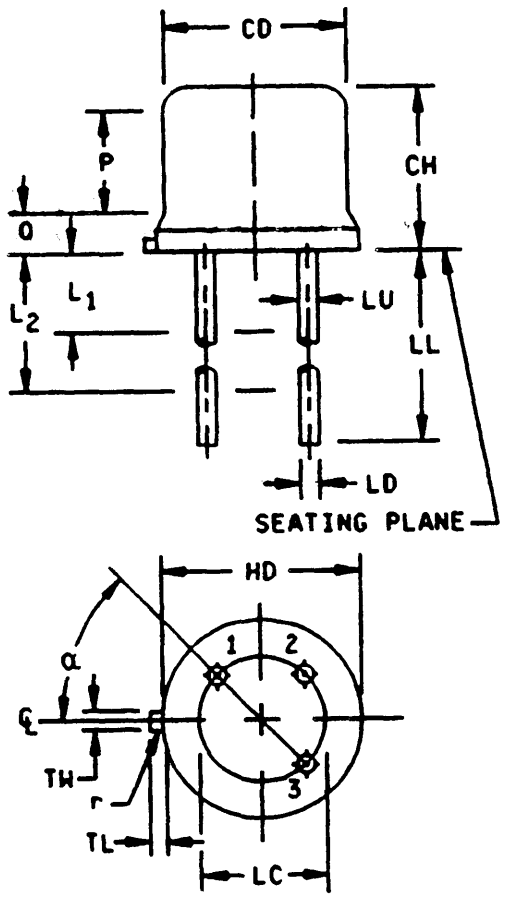
			h_{FE} at V_{CE} = 10 V dc			
	h_{FE1}	h_{FE2}	h_{FE3}	h_{FE4} 1/	h_{FE5} 1/	$R_{\theta JA}$
	I_C = 0.1 mA dc	I_C = 1 mA dc	I_C = 10 mA dc	I_C = 150 mA dc	I_C = 500 mA dc	
	2N2904 2N2905	2N2904 2N2905	2N2904 2N2905	2N2904 2N2905	2N2904 2N2905	°C/mW
Min	20 35	25 50	35 75	40 100	20 30	Max
Max	--- ---	175 450	--- ---	120 300	--- ---	
	2N2904A 2N2905A 2N2904AL 2N2905AL	2N2904A 2N2905A 2N2904AL 2N2905AL	2N2904A 2N2905A 2N2904AL 2N2905AL	2N2904A 2N2905A 2N2904AL 2N2905AL	2N2904A 2N2905A 2N2904AL 2N2905AL	
Min	40 75	40 100	40 100	40 100	40 50	.29
Max	--- ---	175 450	--- ---	120 300	--- ---	

1/ Pulsed (see 4.5.1).

Types 2N2904 2N2904A 2N2904AL 2N2905 2N2905A 2N2905AL	$\|h_{fe}\|$ f = 100 MHz V_{CE} =20 V dc I_C = 50 mA dc	C_{obo} 100 kHz ≤ f ≤ 1 MHz V_{CB} =10 V dc I_E = 0	Switching	
			t_{on} (see figure 2)	t_{off} (see figure 3)
		pF	ns	ns
Min	2.0	---	---	---
Max	---	8	45	300

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Dimensions					
Symbol	Inches		Millimeters		Note
	Min	Max	Min	Max	
CH	.240	.260	6.10	6.60	
LC	.200 TP		5.08 TP		6
LD	.016	.021	0.41	0.53	7,8
LU	.016	.019	0.41	0.48	7,8
HD	.335	.370	8.51	9.40	
CD	.305	.335	7.75	8.51	
Q		.050		1.27	5
TL	.029	.045	0.74	1.14	3,4
TW	.028	.034	0.71	0.86	3
LL	.500	.750	12.70	19.05	7,8,12
L_1		.050		1.27	7,8
L_2	.250		6.35		7,8
r		.010		0.25	10
α	45° TP		45° TP		6



NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 inch (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001, -.000 inch (1.37 +0.03, -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.
12. For "L" suffix devices, dimension LL is 1.50 inches (38.10 mm) minimum, 1.75 inches (44.45 mm) maximum.

FIGURE 1. Physical dimensions (similar to TO-39).

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in figure 1 herein (similar to TO-39).

3.3.1 Lead finish. Lead finish shall be solderable as defined in Mil-S-19500.

3.3.2 Terminal lead length. When terminal lead lengths other than .50 inch (12.7 mm) are required, it shall be stipulated in the contract (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, the following markings may be omitted from the body of the transistor:

a. Country of origin.

b. Manufacturer's identification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
7	hermetic seal	hermetic seal
9	I_{CBO1}, h_{FE2}	Not applicable
10	48 hours minimum	48 hours minimum
11	I_{CBO1}; I_{CES}; h_{FE2}; ΔI_{CBO1} = 100% of initial value or 5 nA dc, whichever is greater. Δh_{FE2} = ±15%	I_{CBO1}, h_{FE4}
12	See 4.3.1 240 hours minimum	See 4.3.1 80 hours minimum
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO1} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE2} = ±15%	Subgroup 2 of table I herein; ΔI_{CBO1} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE4} = ±15%

4.3.1 Power burn-in test conditions. Power burn-in conditions are as follows: T_A = Room ambient as defined in the general requirements of MIL-STD-750, paragraph 4.5;

JANS level (all device types): - - - - - V_{CB} = 10 V dc, P_T = 600 mW
JANTX and JANTXV devices: - - - - - - - V_{CB} = 30 V dc, P_T = 600 mW

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.3.2 Thermal impedance ($Z_{\theta JX}$ measurements) (for qualification only). The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, Method 3131.

a. I_M measurement current - - - - - - - - - - - - 5 mA.
b. I_H forward heating current - - - - - - - - - 200 mA (min).
c. t_H heating time - - - - - - - - - - - - - - 25 - 30 ms.
d. t_{md} measurement delay time - - - - - - - - - 60 µs max.
e. V_{CE} collector-emitter voltage - - - - - - - - 10 V dc

The maximum limit for $Z_{\theta JX}$ under these test conditions are $Z_{\theta JX}$ (max) = 72°C/W.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein. If alternate screening is being performed per 4.6.3.1d and figure 2 of MIL-S-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table IVb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied per 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) of MIL-S-19500. See 4.4.2.2 for JAN, JANTX, and JANTXV group B subgroup testing. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 V dc; P_T = 600 mW at T_A = +25° ±3°C; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on devices shall be permitted.
B5	1027	V_{CB} = 10 V dc; P_T = 600 mW at T_A = 100°C for 96 hours, or T_A = +125°C ±25°C for 96 hours with P_T adjusted according to the chosen T_A to give and average T_J = +275°C.

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV).

Subgroup	Method	Condition
B1	1039	HTRB, test condition A, T_A = 150°C, V_{CB} = 50 V dc, 340 hours, 22 pcs, Acc = 0.
B2	1039	Steady-state life: Test condition B, 1,000 hours, V_{CB} = 10 V dc, P_T = 600 mW at T_A = +25° ±5°C, 45 devices, Acc = 0. No heat sink or forced-air cooling on the devices shall be permitted.
B2	2037	Bond strength, 11 wires, Acc = 0.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. Samples shall be selected from one lot every three months from a die fabrication line that the manufacturer chooses to "certify" for the die design used in products manufactured to this specification.

b. Must be chosen from an inspection lot that has been submitted to and passed group A quality conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B3 and B5 for JANS, and subgroups B1 and B2 for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.2.4 Product acceptable for delivery. The product shall be acceptable for delivery only after the successful completion of all group B testing and shall be comprised of die meeting the following requirements:

a. Manufactured on the same certified die fabrication line as the sample selected for 4.4.2.3.

b. The die lot was manufactured (using start or completion dates, as the manufacturer selects to consistently use) within three months following the lot used for sample selection in paragraph 4.4.2.3. Group B shall not be required more than once for any single die lot.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table III herein.

4.4.3.1 Group C inspection, table V (JANS) of MIL-S-19500.

Subgroup	Method	Condition
C2	2036	Test condition E.
C6	1026	Conditions: 1,000 hours at V_{CB} = 10 V dc; P_T = 600 mW at T_A = +25° ±3°C. No heat sink or forced-air cooling on device shall be permitted.

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4.4.3.2 Group C inspection, JAN, JANTX, and JANTXV. Group C inspection (package related tests) shall be as follows and shall include those package or case-related tests which are performed periodically. Group C test on each package family shall be performed on devices from each six months production of devices (based on inspection lot identification codes) for each assembly location and die attach method. The sample size for each of these subgroups is: 22 devices, c = 0.

Subgroup	Method	Condition
C2	1051	Test condition C.
C2	2036	Test condition E.
C6	---	Not applicable.

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for quality conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.3.4 Product acceptable for delivery. The product which may be accepted for delivery based a successful completion of group C testing shall be product of the particular package type with lot inspection identification codes of the 36 successive weeks beginning with the inspection lot identification code of the successful group C sample (i.e. the inspection lot identification codes accepted for delivery based on successful group C testing will include those sealed during the approximate 10 weeks during which the sample was being screened and tested plus those sealed in the 26 weeks following successful completion of group C testing making a total of 36 successive calendar weeks.) A manufacturer shall not accept product for delivery until after successful completion of the group C testing on product of the applicable package type with an inspection lot identification code less than 35 weeks prior to the inspection identification code of the product being considered for acceptance.

4.4.4 Group D inspection. Quality conformance inspection for hardness assured JANS and JANTXV types shall include the group D tests specified in table II. These tests shall be performed as required in accordance with MIL-S-19500 and MIL-STD-750, method 1019 for total ionizing dose or method 1017 for neutron fluence as applicable.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input capacitance. This test shall be conducted in accordance with method 3240 of MIL-STD-750, except the output capacitor shall be omitted.

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TABLE I. Group A inspection.

Inspection 1/ 2/	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Limits Min	Limits Max	Unit
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage, collector to base	3001	Bias condition D; I_C = 10 µA dc	$V_{(BR)CBO}$	60		V dc
Breakdown voltage, emitter to base	3026	Bias condition D; I_E = 10 µA dc	$V_{(BR)EBO}$	5		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc, Pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N2904, 2N2905				40		V dc
2N2904A, 2N2905A				60		V dc
2N2904AL, 2N2905AL				60		V dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 50 V dc	I_{CBO1}			
2N2904, 2N2905					20	nA dc
2N2904A, 2N2905A					10	nA dc
2N2904AL, 2N2905AL					10	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 3.5 V dc	I_{EBO}		50	nA dc
Collector to emitter cutoff current	3041	Bias condition C	I_{CES}		1	µA dc
2N2904, 2N2905		V_{CE} = 40 V dc				
2N2904A, 2N2904AL		V_{CE} = 60 V dc				
2N2905A, 2N2905AL		V_{CE} = 60 V dc				
Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 0.1 mA dc	h_{FE1}			
2N2904				20		
2N2905				35		
2N2904A, 2N2904AL				40		
2N2905A, 2N2905AL				75		

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Limits Min	Limits Max	Unit
Subgroup 2 - Continued						
Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 1.0 mA dc	h_{FE2}			
2N2904				25	175	
2N2905				50	450	
2N2904A, 2N2904AL				40	175	
2N2905A, 2N2905AL				100	450	
Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 10 mA dc	h_{FE3}			
2N2904				35		
2N2905				75		
2N2904A, 2N2904AL				40		
2N2905A, 2N2905AL				100		
Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 150 mA dc, Pulsed (see 4.5.1)	h_{FE4}			
2N2904, 2N2904A, 2N2904AL				40	120	
2N2905, 2N2905A, 2N2905AL				100	300	
Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 500 mA dc, Pulsed (see 4.5.1)	h_{FE5}			
2N2904				20		
2N2905				30		
2N2904A, 2N2904AL				40		
2N2905A, 2N2905AL				50		
Collector to emitter voltage (saturated)	3071	I_C = 150 mA dc, I_B = 15 mA dc, Pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.4	V dc
Collector to emitter voltage (saturated)	3071	I_C = 500 mA dc, I_B = 50 mA dc, Pulsed (see 4.5.1)	$V_{CE(sat)2}$		1.6	V dc
Base emitter voltage (saturated)	3066	Test condition A, I_C = 150 mA dc, I_B = 15 mA dc, Pulsed (see 4.5.1)	$V_{BE(sat)1}$		1.3	V dc
Base emitter voltage (saturated)	3066	Test condition A, I_C = 500 mA dc, I_B = 50 mA dc, Pulsed (see 4.5.1)	$V_{BE(sat)2}$		2.6	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750 Method	MIL-STD-750 Conditions	Symbol	Limits Min	Limits Max	Unit
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D, V_{CB} = 50 V dc	I_{CBO2}			
2N2904, 2N2905					20	µA dc
2N2904A, 2N2905A					10	µA dc
2N2904AL, 2N2905AL					10	µA dc
Low temperature operation:		T_A = -55°C				
Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 1.0 mA dc	h_{FE6}			
2N2904				15		
2N2905				30		
2N2904A, 2N2904AL				20		
2N2905A, 2N2905AL				50		
Subgroup 4						
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 10 V dc, I_C = 1 mA dc, f = 1 kHz	h_{fe}			
2N2904				25		
2N2905				50		
2N2904A, 2N2904AL				40		
2N2905A, 2N2905AL				100		
Small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 20 V dc, I_C = 50 mA dc, f = 100 MHz	h_{fe}		2.0	
Open circuit output capacitance	3236	V_{CB} = 10 V dc, I_E = 0, 100 kHz ≤ f ≤ 1 MHz	C_{obo}		8	pF
Input capacitance (output open-circuited)	3240	V_{EB} = 2.0 V dc, I_C = 0, 100 kHz ≤ f ≤ 1 MHz, (see 4.5.2)	C_{ibo}		30	pF

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Turn-on time		(See figure 2)	t_{on}		45	ns
Turn-off time		(See figure 3)	t_{off}		300	ns
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plans see MIL-S-19500.
2/ For JANS level, all devices required by the specified LTPD shall be subjected to subgroups 2, 3, and 4 combined.

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TABLE II. Group D inspection and end-point limits.

Inspection	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Neutron irradiation	1017					
Collector to base cutoff current	3036	Bias condition D	I_{CBO1}			
M2N2905A, D2N2905A, R2N2905A, H2N2905A		V_{CB} = 50 V dc			10	nA dc
Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 0.1 mA dc	h_{FE1}			
M2N2905A				61		
D2N2905A				59		
R2N2905A				31		
H2N2905A				5		
Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 1.0 mA dc	h_{FE2}			
M2N2905A				89	450	
D2N2905A				82	450	
R2N2905A				53	450	
H2N2905A				12	450	
Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 10 mA dc, Pulsed (see 4.5.1)	h_{FE3}			
M2N2905A				90		
D2N2905A				87		
R2N2905A				81		
H2N2905A				22		
Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 150 mA dc	h_{FE4}			
M2N2905A				84		
D2N2905A				80		
R2N2905A				58		
H2N2905A				28		
Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 50 mA dc	h_{FE5}			
M2N2905A				45		
D2N2905A				45		
R2N2905A				40		
H2N2905A				20		

See footnote at end of table.

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TABLE II. Group D inspection and end-point limits - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1 - Continued						
Collector to emitter voltage (saturated)	3071	I_C = 150 mA dc, I_B = 15 mA dc, Pulsed (see 4.5.1)	$V_{CE(sat)1}$			
M2N2905A, D2N2905A, R2N2905A, H2N2905A					.44	V dc
Collector to emitter voltage (saturated)	3071	I_C = 500 mA dc, I_B = 50 mA dc, Pulsed (see 4.5.1)	$V_{CE(sat)2}$			
M2N2905A, D2N2905A, R2N2905A, H2N2905A					1.75	V dc
Subgroup 2						
Steady-state total dose irradiation	1019					
Collector to base cutoff current	3036	Bias condition D	I_{CBO1}			
M2N2905A, D2N2905A, R2N2905A, H2N2905A		V_{CB} = 50 V dc			10	nA dc
Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 0.1 mA dc	h_{FE1}			
M2N2905A, D2N2905A				75		nA dc
R2N2905A				25		
H2N2905A				5		
Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 1.0 mA dc	h_{FE2}			
M2N2905A				89	450	
D2N2905A				82	450	
R2N2905A				43	450	
H2N2905A				13	450	
Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 10 mA dc	h_{FE3}			
M2N2905A				90		
D2N2905A				87		
R2N2905A				57		
H2N2905A				32		

See footnote at end of table.

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TABLE II. Group D inspection and end-point limits - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 150 mA dc, Pulsed (see 4.5.1)	h_{FE4}			
M2N2905A				84		
D2N2905A				81		
R2N2905A				58		
H2N2905A				48		
Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 500 mA dc, Pulsed (see 4.5.1)	h_{FE5}			
M2N2905A, D2N2905A				50		
R2N2905A				40		
H2N2905A				24		
Collector to emitter voltage (saturated)	3071	I_C = 150 mA dc, I_B = 15 mA dc, Pulsed (see 4.5.1)	$V_{CE(sat)1}$			
M2N2905A, D2N2905A					.44	V dc
R2N2905A					.44	
H2N2905A					.50	
Collector to emitter voltage (saturated)	3071	I_C = 500 mA dc, I_B = 50 mA dc, Pulsed (see 4.5.1)	$V_{CE(sat)2}$			
M2N2905A, D2N2905A					1.75	V dc
R2N2905A					2.2	
H2N2905A					2.5	

1/ For sampling plan see MIL-S-19500.

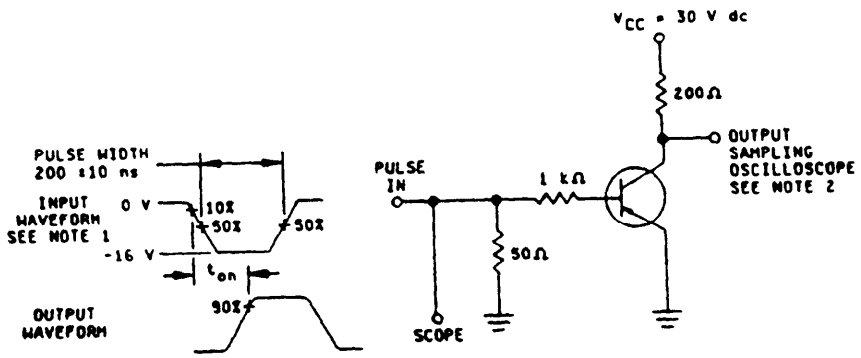
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TABLE III. Groups B and C electrical measurements.2/, 3/, 4/, 5/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1	Collector to base cutoff current	3036	Bias condition D, V_{CB} = 50 V dc	I_{CBO1}			
	2N2904, 2N2905					20	nA dc
	2N2904A, 2N2905A					10	nA dc
	2N2904AL, 2N2905AL						
2	Collector to base cutoff current	3036	Bias condition D, V_{CB} = 50 V dc	I_{CBO2}			
	2N2904, 2N2905					40	nA dc
	2N2904A, 2N2905A					20	nA dc
	2N2904AL, 2N2905AL					20	nA dc
3	Emitter to base cutoff current	3061	Bias condition D, V_{EB} = 3.5 V dc	I_{EBO}		50	nA dc
4	Collector to emitter voltage (saturated)	3071	I_C = 150 mA dc, I_B = 15 mA dc, Pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.4	V dc
5	Base emitter voltage (saturated)	3066	Test condition A, I_C = 150 mA dc, I_B = 15 mA dc, Pulsed (see 4.5.1).	$V_{BE(sat)1}$		1.3	V dc
6	Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 1.0 mA dc	h_{FE2}			
	2N2904				25	175	
	2N2905				50	450	
	2N2904A, 2N2904AL				40	175	
	2N2905A, 2N2905AL				100	450	
7	Collector to base cutoff current	3036	Bias condition D, V_{CB} = 50 V dc	ΔI_{CBO1} 1/		100% of initial reading or 8 nA dc whichever is greater.	
8	Forward current transfer ratio	3076	V_{CE} = 10 V dc, I_C = 1.0 mA dc, Pulsed (see 4.5.1).	Δh_{FE2} 1/		±25% change from initial reading.	

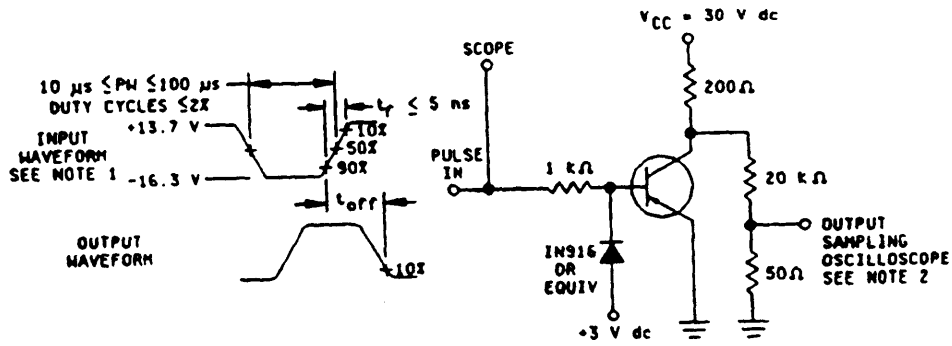
1/ Devices which exceed the group A limits for this test shall not be accepted.
2/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:
   a. Subgroup 3, see table III herein, steps 1, 3, 4, 5, and 6.
   b. Subgroups 4 and 5, see table III herein, steps 2, 3, 4, 5, and 8.
3/ The electrical measurements for table IVb (JAN, JANTX, and JANTXV) are as follows:
   a. Subgroups 1 and 2, see table III herein, steps 1, 6, 7, and 8.
4/ The electrical measurements for group C (table V, of MIL-S-19500) (for JANS) are as follows:
   a. Subgroup 2 and 3, see table III herein, steps 1, 3, 4, 5, and 6.
   b. Subgroup 6, see table III herein, steps 2, 3, 4, 5, 6, 7, and 8.
5/ The electrical measurements for group C (table V, of MIL-S-19500) (for JAN, JANTX, and JANTXV) are as follows:
   a. Subgroup 2 and 3, see table III herein, steps 1 and 6.

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NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50 Ω.
2. Sampling oscilloscope: Z_{in} ≥ 100 kΩ, C_{in} ≤ 12 pF, rise time ≤ 5 ns.

FIGURE 2. Saturated turn-on switching time test circuit.



NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50 Ω.
2. Sampling oscilloscope: Z_{in} ≥ 100 kΩ, C_{in} ≤ 12 pF, rise time ≤ 5 ns.

FIGURE 3. Saturated turn-off switching time test circuit.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Lead finish (see 3.3.1).

d. Lead length, if other than 0.5 inch (see 3.3.2).

e. Type desisgnation, quality assurance level, and radiation level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 13, 15, 19, 85, 99
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/290E	2. DOCUMENT DATE (YYMMDD) 950517

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, SWITCHING TYPES 2N2904, 2N2904A, 2N2904AL, 2N2905, 2N2905A, AND 2N2905AL, JAN, JANTX, JANTXV, JANS, JANTXVM, JANTXVD, JANTXVR, JANTXVH, JANSM, JANSD, JANSR, AND JANSH

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME DIRECTOR, US ARMY RESEARCH LAB/PSD	b. TELEPHONE (Include Area Code) (1) Commercial: 908-427-3441/3148 (2) AUTOVON: 987-3441/3148
c. ADDRESS (Include Zip Code) ATTN: AMSRL-PS-DC FORT MONMOUTH, NJ 07703-5601	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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